8X

SECTION A: CHOOSE THE SINGLE BEST OPTION [50 marks]

CHOOSE THE SINGLE BEST OPTION. EACH QUESTIONS CARRIES 1 MARK

1. Iron is present in all of the following EXCEPT:
   a) Heme
   b) Bilirubin
   c) Myoglobin
   d) Ferritin

2. Chronic blood loss in a person can result in which type of anemia?
   a) Hemolytic anemia
   b) Hypochromic anemia
   c) Megaloblastic anemia
   d) Aplastic anemia

3. The protein present in the vascular endothelium that combines with thrombin and prevents intravascular clotting?
   a) Plasminogen
   b) Thrombomodulin
   c) Thrombosthenin
   d) Heparin

4. The aggregation of platelets is inhibited by:
   a) ADP
   b) Prostacyclin
   c) Thrombosthenin
   d) Thromboxane A 2

5. The activation of which clotting factor is crucial for the extrinsic mechanism of blood coagulation?
   a) V
   b) VII
   c) VIII
   d) X

6. Regarding breakdown of erythrocytes in the body, all are true EXCEPT:
   a) Takes place in the reticulo-endothelial system
   b) Yields iron, most of which is retained for further use
   c) Yields bilirubin which is carried by plasma protein to the liver
   d) Is required for the synthesis of bile salts.

7. Iron deficiency
   a) Frequently follows persistent loss of blood from the body
   b) Is more common in men than in women
   c) May cause anemia by inhibiting the rate of multiplication of RBC stem cells
   d) May cause large pale erythrocytes to appear in peripheral blood

8. The hematocrit, all are true EXCEPT:
   a) May be obtained by centrifugation of blood
   b) May be calculated by multiplying the mean cell volume by the red cell count
   c) Rises in a patient who sustains widespread burns
   d) Rises in macrocytic megaloblastic anemia such as pernicious (B12 deficiency) anemia

9. Red cell formation is increased, all are true EXCEPT:
   a) By giving vitamin B12 injections to healthy people on a normal diet
   b) In patients with hemolytic anemia
   c) By giving injections of erythropoietin to patients with removed kidneys
   d) In patients who have a raised blood reticulocyte count

10. Deficiency of factor VIII
    a) Increases the clotting time
    b) Is due to an abnormal gene on the Y chromosome
    c) Causes small (petechial) hemorrhages into the skin to cause purpura
    d) Affects the extrinsic, rather than the intrinsic, pathway for blood coagulation

11. A patient has a congenital deficiency in factor XIII (fibrin-stabilizing factor). What would analysis of his blood reveal?
    a) Prolonged prothrombin time
    b) Prolonged whole blood clotting time
    c) Prolonged partial thromboplastin time
    d) Easily breakable clot

12. Hematocrit is increased in:
    a) Dehydration
    b) Pregnancy
    c) Chronic renal failure
    d) Thalassemia

13. Reticulocytosis occurs in:
    a) Bone marrow stimulation
    b) Vitamin B12 deficiency
    c) Iron deficiency
    d) X ray radiation

14. Hemoglobin structure is abnormal in:
    a) Thalassemia
    b) Pernicious anemia
    c) Iron deficiency anemia
    d) Polycythemia

15. Mean corpuscular volume is increased in:
   a) Iron deficiency
   b) Acute blood loss
   c) Hereditary spherocytosis
   d) Vitamin B12 deficiency

16. The immediate precursor of RBC in its development is:
   a) Stem cell
   b) Proerythroblast
   c) Reticulocyte
   d) Late normoblast

17. During intrauterine life, formation of the red blood cells begins in:
   a) Bone marrow
   b) Liver
   c) Mesoderm of the yolk sac
   d) Spleen

18. In a patient having vitamin K deficiency, there is:
   a) Normal clotting time
   b) Prolonged bleeding time
   c) Prolonged prothrombin time
   d) Thrombocytopenia

19. Blood platelets assist in arresting bleeding by the following EXCEPT:
   a) Releasing factors promoting blood clotting
   b) Adhering together to form plugs when exposed to collagen
   c) Liberating high concentrations of calcium
   d) Releasing factors causing vasoconstriction

20. Antibodies (agglutinins) of the A and B red cell antigens (agglutinogens)
   a) Are present in fetal plasma
   b) Cause hemolysis of RBCs containing the A and B antigens when added to a suspension of red cells in saline
   c) Do not normally cross the placental barrier
   d) Have one binding site (monovalent)

21. A woman whose blood type is A, Rh positive, and a man whose blood type is B, Rh positive, come to the clinic with a 3-year-old girl whose blood type is O, Rh negative. What can be said about the relationship of these two adults to this child?
   a) The woman can be the child's natural mother, but the man cannot be the natural father
   b) The man can be the child's natural father, but the woman cannot be the natural mother
   c) Neither adult can be the natural parent of this child
   d) This couple can be the natural parents of this child

22. What is the appropriate treatment for an infant born with severe erythroblastosis fetalis?
   a) Passive immunization with anti-Rh(D) immunoglobulin
   b) Immunization with Rh(D) antigen
   c) Exchange transfusion with Rh(D)-positive blood
   d) Exchange transfusion with Rh(D)-negative blood

23. Which agent is not effective as an in vitro anticoagulant?
   a) Heparin
   b) Warfarin (Coumadin)
   c) Ethylenediamine tetraacetic acid (EDTA) (calcium chelator)
   d) Sodium citrate (calcium chelator)

24. What would most likely be used for prophylaxis of an ischemic heart attack?
   a) Heparin
   b) Warfarin
   c) Aspirin
   d) Streptokinase

25. The enzyme that ultimately lyses fibrin is:
   a) Plasminogen
   b) TPA
   c) Urokinase
   d) Plasmin

26. The test that screens the extrinsic pathway is:
   a) Prothrombin time (PT)
   b) Activated partial thromboplastin time (APTT)
   c) Thrombin time
   d) Bleeding time

27. Bleeding time is determined by nicking the skin superficially with a scalpel blade and measuring the time required for hemostasis. It will be markedly abnormal (prolonged) in a person who has which of the following?
   a) Anemia
   b) Vitamin K deficiency
   c) Thrombocytopenia
   d) Hemophilia

28. The lower the total amount of iron stored in the body,
   a) The higher the rate of red blood cell formation
   b) The higher the absorption of ingested iron in the intestine
   c) The lower the level of transferrin in the blood
   d) The higher the level of ferritin in the liver

29. In which choice are the proteins listed in the order in which they are activated?
   a) Prothrombin activator, thrombin, fibrin
   b) Fibrin, thrombin, prothrombin activator
   c) Thrombin, fibrin, prothrombin activator
   d) Thrombin, prothrombin activator, fibrin

30. When red blood cells are worn out, part of their components are recycled while others are disposed. Select the incorrect statement about destruction of red blood cells.
   a) The greenish pigment, biliverdin, is recycled to the bone marrow
   b) Iron is carried to the bone marrow by a protein called transferrin
   c) Biliverdin and bilirubin impart color to bile
   d) Macrophages in the liver and spleen destroy worn out red blood cells

31. While attending an emergency care patient, which of the following isotonic fluid is used for IV infusion ?
   a) 1.9% Nacl
   b) 9% Nacl
   c) 5% Dextrose
   d) Ringer's Glucose

32. The Sodium potassium pump prevent cell from swelling by pumping:
   a) 3 Na+ inside
   b) 3Na+ outside
   c) 3K+ inside
   d) 3K+ outside

33. An example of an integral protein in a cell membrane:
   a) Spectrin
   b) Ankyrin
   c) Actin
   d) Band 3

34. The semi-permeable function of the plasma membrane helps to
   a) Fight against cellular infection
   b) Establish the resting membrane potential
   c) Protects cell from toxic substances
   d) Maintain the size of the cell

35. All of the following are not examples of passive transport mechanism across cell membrane except:
   a) Calcium pump
   b) Facilitated transport
   c) Primary active transport
   d) Endocytosis

36. An example of active transport mechanisms across cell membrane include:
   a) Endocytosis
   b) Osmosis
   c) Filtration
   d) Facilitated transport

37. Skeletal muscle number is fixed at time of birth in humans. Which of the following hormones stimulate muscle hypertrophy?
   a) Estrogen
   b) Growth hormone
   c) Oestrone
   d) Adrenaline

38. Functional significance of sarcoplasmic reticulum
   a) Membranous sacs which encircles each myofibril
   b) Stores and release calcium ions (Ca++)
   c) Release of Na+ triggers muscle contraction
   d) Interacts with Troponin C

39. Regarding contractile filaments
   a) Function in the non contractile process
   b) Three types of filaments (Thick, medium and Thin)
   c) There are one thick filaments for every two thin filaments
   d) There are two thin filaments for every thick filament

40. Regarding Sarcomeres:
   a) These are compartments of arranged fibres
   b) Generate action potential
   c) Basic functional unit of a myofibril
   d) Store and release Ca++

41. Regarding myosin filaments
   a) They are thin filaments
   b) Functions as a regulatory protein which can achieve motion
   c) Convert ATP to glucose of motion
   d) Each myosin molecule protrude outward as a head

42. The muscle protein which covers the active sites on the actin filament at rest is:
   a) Actin
   b) Myosin
   c) Tropomyosin
   d) Troponin-T

43. The total strength of a contraction depends on the:
   a) Motor neurons
   b) Less motor units
   c) Motor units activated
   d) Stimulus strength

44. During Relaxation period of a simple muscle twitch:
a) Na+ is transported into the SR
b) Myosin sticks to sarcolemma
c) Myosin-binding sites are covered by tropomyosin
d) Power stroke of myosin occurs

45. Regarding Repolarisation:
a) It is the decrease in cell electronegativity
b) It is the increase in cell electropositivity
c) It is the decrease in cell electropositivity inside membrane
d) It is the decrease in cell electronegativity outside

46. An action potential in a nerve fibre:
a) Has amplitude which varies directly with the strength of the stimulus
b) Induces graded electrical currents in adjacent segments of the fibre
c) Is associated with a transient decrease in membrane permeability to potassium
d) Is associated with a transient increase in membrane permeability to sodium

47. About the Na+/K+ pump of cell membranes, the following hold true;
a) All cells have the Na+K+ pump that pumps in Na+ and takes out K+ of the cell actively
b) 2 potassium ions are pumped out
c) Breaks down ATP to power ion transport
d) The Na+K+ pump does not require it beta subunit to operate

48. Neurons that stimulate skeletal muscle to contract are labelled as
a) Somatic sensory neurons
b) Somatic motor neurons
c) Autonomic motor neurons
d) Autonomic sensory neurons

49. Skeletal muscle have no capacity for regeneration. Which of these cells retain the capacity to regenerate only the damaged muscle fibers?
a) Myosin-1cells
b) Satellite cells
c) Endothelial cells
d) Basal laminar cells

50. Function of Contractile proteins inside myofibrils is
a) Switch the contraction process prolonged
b) Switch the contraction process off
c) Generate force during contraction
d) Align the thick and thin filaments properly

SECTION B: STRUCTURED AND SHORT ANSWER QUESTIONS [50 marks]

Answer all the questions in the provided spaces within the question paper

Answer the first 25 items with:

A. Increase (I)

B. Decrease (D)

C. Stay the same (S)

19

1. An increase in plasma levels of erythropoietin will cause blood viscosity to: Same (S)
2. The activation of plasminogen will cause the size of a blood clot to: Decrease (D)
3. A decline in plasma albumin levels would cause plasma osmotic pressure to: Decrease (D)
4. Aspirin causes the likelihood of platelet activation to: Decrease (D)
5. A reduced ability to produce thrombin would cause the time required for blood clot formation to: Decrease (D)
6. A decrease in white blood cell count would cause the likelihood of an infection to: Increase (I)
7. During the body's response to an acute bacterial infection, you would expect the neutrophil count to: Decrease (D)
8. During the vascular spasm phase of hemostasis, the diameter of the affected blood vessel will: Decrease (D)
9. If the liver was unable to produce normal quantities of plasma proteins, plasma osmotic pressure would: Decrease (D)
10. As plasma osmotic pressure decreases, interstitial fluid formation will: Increase (I)
11. During an acute bacterial infection, blood neutrophil count will: Decrease (D)
12. Prostacyclin causes the likelihood of platelet plug formation to: Decrease (D)
13. Plasmin causes plasma fibrin levels to: Decrease (D)
14. In response to infection, plasma leukocyte count would be expected to: Increase (I)
15. In response to damage to the wall of a blood vessel, the level of contraction of that blood vessel's smooth muscle will: Increase (I)
16. Urokinase is an enzyme that activates plasminogen. Thus, urokinase causes the likelihood of clot formation to: Decrease (D)
17. A mismatched transfusion of Type AB blood into an individual with Type O blood would make plasma bilirubin to: Increase (I)

18. During an infection the size of the buffy coat will: Increase (I)

19. During a severe allergic reaction, the number of eosinophils in the body would: Increase (I)

20. If plasmin was activated prior to clot formation, the time it takes for clot formation to occur would: Decrease (D)

21. During leukopenia the body's ability to prevent bacterial infection will: Decrease (

22. A decrease in intrinsic factor production would cause the blood's oxygen carrying capacity to: Same (S)

23. Aplastic anemia would cause the body's WBC count to: Decrease (D)

24. Infection with a parasitic fluke would cause the number of granulocytes in the body to: Increase (I)

25. Large doses of antibiotics would cause the ability of the body to form thrombin to: Same (S)

26. Michael Mumba, LeBron James, Kareem Abdul-Jabbar are men who have changed the history of basketball. What would happen to membrane electricity, if one of these players had increased potassium in the ECF. Outline the physiology that would be disrupted at plasma membrane level. **[5 marks]**

IF K^+ was incrase in the ECF And there would be an in movement of water out of the cell to the ECF causing Edema. Edema is the accumulation of fluids in the interstitial fluid. This can occur due to movement of water from the cells to the ECF fluid by osmosis and from the blood to the ECF.

27. What would happen if one of these players had electrolyte derangements during the game? Outline the physiological balance and distribution of normal electrolytes in body fluids. [5 marks]

Electrolytes have an important role to play in fluidity of the extracellular fluids and intracellular fluids, if there is a slight balance, either water will move in or out of the cell, hence it may cause hemolysis ~~the~~ in terms of RBC's, or water would move in the blood in the cases when electrolytes where too be low in the blood

2

28. Compare and contrast the applied CLINICAL significance of osmolarity and osmolality. [5 marks]

Osmolarity is the measure of osmoles per litre of a solute, it is measured in mol/L or M while osmolality is the measure of osmoles per kg of the solvent it is measured in mol/kg or M. Osmolality is more accurate as compared to osmolarity because mass does not change while volume can change under the influence of ~~tempreture~~ among others. Osmolarity is more difficant to measure as compared to osmolality.

1½

29. Discuss succinctly the types of IV fluids that are needed to correct his electrolyte imbalance based on tonicity. [5 marks]

~~If there is a~~ Isotonic fluids are fluids that have same concentration of salts as that of plasma, an example is 5%

1½

Dextrose, Normal Saline and Ringers Lactate.
hypotonic fluids have lower concentration of electrolytes than plasma so water moves ~~in~~ of the cell and hypertonic fluids have a concentration higher than that of plasma so water moves out causing crenation

30. COMPARE and CONTRAST the physiology of gap junctions and tight junctions with needed EXAMPLES.

[5 marks]

~~Gamp~~ Gap junctions are important for cell-to cell communication, an example are ~~gap~~ gap junctions found in the heart, gap junctions are made up of Connexons and Connexins.
2½
Tight junctions are used as diffusion barriers they are made up of occludins and claudins, an example are the tight junctions found ~~in~~ in the stomach

SECTION C: SCENARIO QUESTION

18

[50 marks]

Answer all the questions in the spaces provided within the question paper

PLEASE NOTE ONE MARK IS NOT EQUIVALENT TO A SUMMARIZED ANSWER.

1. A 40-year-old woman visits the clinic complaining of fatigue. She had recently been treated for an infection. Her laboratory values are as follows: red blood cell (RBC. count, 1.8×10^6 /µl; hemoglobin (HB., 5.2 g/dL; hematocrit (Hct), 15%; white blood cell (WBC. count, 7.6×10^3 /µl; platelet count, 320,000/µl; mean corpuscular volume (MCV), 75 fL; MCH=26 pg; MCHC= 28 g/dL and reticulocyte count, 20%.

a) What do the Hb, MCV, MCH, and MCHC values indicate about this patient's red blood cells? [3 marks]

Hb → Hemoglobin
MCV → Mean Corpuscular (cell) volume
MCH → Mean Corpuscular (cell) hemoglobin
MCHC → Mean Corpuscular (cell) hemoglobin Concentration.

0.5

b) Mention four factors that could be responsible for the values of Hb and Hct in the patient above [2 marks]

The decrease in the hematocrit may have been due to blood loss and the decrease in the hemoglobin content is due to the low amounts of red blood cells in the body.

00

c) What role do reticulocytes play in blood physiology, and why might the elevated count be observed in this patient? [2 marks]

The role of reticulocytes is to show how the functionality of the bone marrow is. The Increase in this patients reticulocyte is due to the low levels of ~~RB~~ RBC's in the body, hence the bod is trying to Compensate for the lost RBC's.

02

d) What is the consequence of the RBC count in the patient and how does the body respond to that? [2 marks]

~~The role of~~ The Consequence of the low RBC Count will lead to a low oxygen Supply to the tissues which causes hypoxia. The kidneys will be stimulated by hypoxia to produce erythropoietin. Erythropoietin stimulates erythropoiesis (RBC Formation).

02

e) Do you think this patient will develop a bleeding condition as a result of her platelet count? Explain.

[1 mark]

No, because her platelet count (320,000/uL) is within the normal range of ~~platelet~~ count which is 150,000 - 450,000/uL.

01

2. Several days after returning home from the US, a 70-year old woman awoke with swelling and pain in her right leg, which was blue. She immediately went to the emergency department, where examination showed an extensive deep vein thrombosis involving the femoral and iliac veins on the right side. She has been undergoing stable and successful anticoagulation with warfarin for recurrent deep vein thrombosis (DVT) was treated for pneumonia. Her prothrombin time was quite prolonged.

a) Describe the mechanism of action of Warfarin

[3 marks]

Warfarin inhibits vitamin K competitively. If vitamin K is inhibited there will be no conversion of glutamic acid to gammaglutamic acids. Factors II, VII, IX and X depends on this conversion for their release into the blood stream from the liver. Hence there will be no release of factors II, VII, IX and X.

03

b) What could be the reason for the prolonged prothrombin time in this patient?

[1 mark]

This is due to the lack of factor VII which is need for the extrinsic pathway.

00

3. As per UN reports, nearly 40% of people in Tigray region of Ethiopia are currently suffering from lack of food to eat.

a) How does this affect body fluid balance and homeostasis in terms of malabsorption diarrhoea?

[5 marks]

IF people are not eating enough there will be less electrolytes in the body ~~due to insufficient~~, IF electrolytes are low in the intestines, electrolytes will start moving from the enterocytes to the intestines, this causes a shift in the fluids due to high electrolytes in the intestinal lumen.

1½

b) What lab tests can assist in patient assessment? Highlight the normal values in mEq/L

[5 marks]

0

c) Elucidate the role of plasma membrane and channels in maintaining body fluid electrolytes.

[5 marks]

The plasma membrane maintains body fluid electrolytes by allowing certain substances to pass through and certain substances not to pass through. Channels allows large and polar substances to pass

2

Sec B = 08

c) Draw the extrinsic pathway of blood clotting [3 marks]

- factor V is activated by thrombin.

03

Damaged Endothelial cells
↓
factor III
↓
VII ⟶ VIIa
↓ ⟶ in the presence of - Ca2+ - platelete fact 3 - factor III
X ⟶ Xa
In the presence of - Ca 2+ - platelet fact 3 - factor V
↓
Prothrombin ⟶ Thrombin
↓
fibrinogen ⟶ fibrin
↑ (makes it firm and tight)
XIII

d) How does prolonged immobility, such as during a long flight, contribute to the risk of developing intravascular clotting? [2 marks]

02

When ~~Someth~~ Someone is at rest, the blood flow velocity decreases as compared to someone who is active. This leads to stugnat hypoxia (due to slugish flow of blood). This will increase the chances of platelet aggregation.

e) State 4 ways by which the intact endothelium prevents clotting [4 marks]

03

- The balance between Thromboxan A_2 and prostacyclin
- High speed of blood flow
- presence of Heparin Sulfate which activates antithrombin III
- presence of thrombomodulin which binds to thrombin and activates protein C

f) Is blotting clotting in this patient by the intrinsic or extrinsic pathway? Explain. [2 marks]

02

It is by the intrinsic pathway because there was no injure to a blood vessel, meaning there was no release of platelet factor III.

through the cell membrane ~~and out of~~ In and out, this balances the amount of electrolytes inside and out of the cell.

d) Under normal cell conditions, Is Donnan effect hazardous? How is this prevented inside the human body cells? DISCUSS this physiological concept [5 marks]

Yes donnan effect is hazardous in that it causes swelling of the cell as chloride ions enters the cell, water also follows. This is ~~is~~ prevented by the Na^+/K^+ pump which pumps in 2 K^+ ions and 3 Na^+ ions out side the. The pumping ~~of~~ out of 3 Na^+ ions will cause Cl^- ions to follow down the electrical potential, this will also cause ~~the~~ water molecules to move out of the cell by osmosis.

2

THE END